Algebra: Mixed Word Problems

Core Concepts with Algebra II Extensions

Name: _____ Date: _____

Instructions

Allow 30 minutes for Problems 1–9 and 20 additional minutes for Problems 10–12. A basic calculator is allowed. Define variables, write an equation or inequality, and show your reasoning. Include units and interpret answers in context; reject any solutions that do not fit the situation. Give exact answers unless rounding is explicitly requested. Give money amounts to the nearest cent when needed. Use the space provided or attach extra paper.

1. A club makes a display using red and blue tiles in the ratio $3 : 5$. There are 64 tiles altogether.

   (a) How many tiles of each color are there?

   (b) The club adds 16 blue tiles and no red tiles. What is the new ratio of red tiles to blue tiles, in simplest form?

2. A school play sells adult tickets for \$9 and student tickets for \$5. A total of 86 tickets brings in \$626. How many tickets of each type were sold?

3. A club has \$485 for a field trip. Transportation costs a flat \$125, and admission costs \$8 per student. At least 30 students must attend. There are no other costs. Describe all possible numbers of students who can attend without exceeding the budget. State your answer both as an inequality and in words.

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4. Pump A can fill an empty tank in 6 hours, and Pump B can fill the same tank in 9 hours. Each pump operates at a constant rate, and no water leaves the tank.

   (a) How long will the pumps take to fill an empty tank together?

   (b) Starting with an empty tank, Pump A runs alone for 2 hours. Pump B then joins it. How much *additional* time is needed to finish filling the tank?

   Express both answers in hours and minutes.

5. A map has a scale of $1 : 25{,}000$: 1 cm on the map represents 25,000 cm on the ground. A trail measures 7.2 cm on the map. A hiker walks the entire trail at a constant speed of 4.5 km/h, without stopping. Find the actual trail length in kilometers and the walking time in minutes. Use 100 cm = 1 m and 1,000 m = 1 km.

6. A robotics competition defines a performance score by

$$Q = \frac{120n}{t(e+1)},$$

   where n is the number of completed deliveries, t is the positive elapsed time in minutes, and e is the number of errors.

   (a) Find the score for 30 deliveries in 5 minutes with 1 error.

   (b) A robot completes 30 deliveries with 2 errors and earns a score of 300. How many minutes did it take?

   (c) If the deliveries and time in part (a) stay the same but the errors increase from 1 to 3, by what factor does the score change?

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7. Six students record the following reading times, in minutes:

$$12, \quad 14, \quad 14, \quad 16, \quad 19, \quad x.$$

The mean is 17 minutes.

(a) Find x and the median reading time.

(b) The student whose time was x corrects it to 57 minutes. Find the new mean and median.

(c) Which measure better describes a typical reading time after the correction? Justify your choice using the data.

8. A lab needs 12 liters of a solution that is 30% concentrate by volume. It has only a 20% solution and a 35% solution of the same concentrate. Assume volumes add when mixed. How many liters of each solution should be combined?

9. A supplier charges a fixed delivery fee plus the same price for each kit. The table shows three complete orders; there are no taxes or other charges.

Number of kits	2	5	9
Total cost (dollars)	19	37	61

(a) Find the cost per kit and the delivery fee. Write a formula for the total cost C of an order of n kits.

(b) Predict the cost of 20 kits.

(c) What is the greatest number of kits that can be ordered with \$100, including delivery?

Algebra: Mixed Word Problems — Continued

10. Two printers work at constant rates. The faster printer takes 3 hours less than the slower printer to complete the same printing job alone. Working together, they complete one such job in 2 hours. How long would each printer take to complete the job alone? Show why any rejected algebraic solution cannot describe the situation.

11. A rectangular garden measures 20 meters by 14 meters. A path of uniform width is built *inside* the garden along all four edges, leaving a smaller rectangle for planting. The planting area is 160 square meters. Find the width of the path and the dimensions of the planting rectangle. Explain any restriction on the path width.

12. At a price of \$20 per sweatshirt, a club expects to sell 80 sweatshirts. For each \$1 increase in price, it expects to sell 2 fewer sweatshirts. The club pays \$6 for each sweatshirt sold and has no other costs. Assume the prediction is exact. Prices may only increase from \$20 in whole-dollar steps, and predicted sales must be nonnegative.

    (a) Let x be the number of \$1 price increases. Write a formula for the club's total profit and state the possible values of x.

    (b) What selling price produces the greatest profit? Find that profit and the predicted number of sweatshirts sold.

    (c) Find all allowed selling prices for which the predicted profit is at least \$1,400.